

Portfolio value-at-risk forecasting with GA-based extreme value theory

Ping-Chen Lin^{a,*}, Po-Chang Ko^b

^a Institute of Finance and Information, National Kaohsiung University of Applied Sciences, 415 Chien Kung Road, Kaohsiung 80, Kaohsiung 807, Taiwan

^b Department of Information Management, National Kaohsiung University of Applied Sciences, 415 Chien Kung Road, Kaohsiung 80, Kaohsiung 807, Taiwan

Abstract

Value-at-risk (VaR) has become a popular risk measure since it was adopted by the Bank for International Settlements and US regulatory agencies in 1988. The VaR concept has also been further extended to the portfolio value-at-risk (PVaR) measure used for managing risks and returns under a multiple-asset portfolio. Precise prediction of PVaR provides better evaluation criteria in areas such as investment decision-making and risk management. The two issues concerned with portfolio risk are efficient set selection and volatility forecasting. Most of the statistical portfolio selection models are based on linear functions under specific assumptions. Due to the fat-tailed distribution in most real financial time-series data, extreme value theory (EVT) is powerful in determining the VaR of a portfolio by concentrating on estimating the shape of the fat-tailed probability distribution. However, using EVT to evaluate the portfolio's volatility is very difficult, because each asset within the portfolio has its own distinct peak threshold value. This study introduces an evolutionary portfolio volatility forecasting model to optimize portfolios under their maximum expected returns subject to a risk constraint. We use a genetic algorithm (GA) to extract the best portfolio set and most suitable peak threshold in order to estimate the portfolio's VaR by means of EVT.

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1. Introduction

Volatility forecasting has become an important input in the creation of many investments, options, and portfolios (Poon et al., 2003). Portfolio managers and investors have certain risk threshold levels that they can bear. Portfolio volatility forecasting involves two key issues: efficient set selection and value-at-risk (VaR) measurement. Efficient set selection aims to allocate assets by maximizing the expected risk premium per unit of risk. The earlier portfolio selection methodologies were the mean-variance approach pioneered by Markowitz (1952), quadratic programming (Crama & Schyns, 2003), goal programming (Arenas, Bilbao, & Rodríguez, 2001), and dynamic pro-

gramming, (Yiu, 2004), etc. However, most of them are based on linear functions under specific assumptions. Besides, VaR has become a popular risk measure since it was recommended and adopted by the Bank for International Settlements and US regulatory agencies in 1988. Regulators are very concerned about the capital of bank incurring the maximum possible loss if they underestimated the systemic risk of business during specified (extraordinary) period, such as stock market crashes or foreign exchange crises, and, etc. The VaR concept has been further extended to the portfolio value-at-risk (PVaR) measure used to evaluate the maximum potential loss of a portfolio with a given probability over a specified period (Manganelli & Engle, 2001). For risk management, investment, and regulatory purposes, PVaR is also used to predict the probability of an extreme movement in the value of a portfolio (Campbell, Huisman, & Koedijk, 2001).

* Corresponding author. Tel.: +886 7 3814526; fax: +886 7 3831544.
E-mail address: lety@cc.kuas.edu.tw (P.-C. Lin).

The extreme movements are related to the tails of the distribution, which is especially important because of the fat-tailed characteristics in most financial time series data. Extreme value theory (EVT) focuses directly on the tails of distributions and therefore potentially provides better estimates of risk (Bystrom, 2004).

PVaR is defined as the maximum loss on a portfolio that can be expected with a certain level of confidence over a certain interval of time. The variance–covariance and historical simulation approaches are the most common methods used for VaR forecasting. The variance–covariance method was introduced by Riskmetrics, (Morgan, 1996) that estimated the variances and co-variances of asset return for calculating VaR. The earlier well-known model is simple moving average (SMA). Its important assumption is that the stock (portfolio) returns are normally distributed, which indicates the loss would be equaled or exceeded x percent of the time. The exponential weighted moving average (EWMA) method is another famous model used by Morgan later which allows volatility clustering in order to better fit modelling purposes in presence of leptokurtosis. Unfortunately, the variance–covariance method tends to underestimate the VaR since the normality assumption of the standardized residuals seems not to be consistent for the real behavior of financial returns (Danielsson & de Vries, 1997a). Due to the right tail is biased downwards or left tail is biased upwards for high quantiles of a fat-tailed distribution, the VaR will be further underestimated (Duffie & Pan, 1997). The historical simulation (HS) is another common method to forecast VaR. It drastically simplifies the calculation process of VaR, since it does not require any explicit assumptions on the distribution of portfolio returns. But, the estimation process is not reliable since HS are calculated from limited observations (Danielsson & de Vries, 1997b). It is not possible to obtain any quantile estimates above the highest observed quantile (Barone-Adesi & Kostas, 2000).

The central limit theory treats fluctuations of cumulative sums. In contrast with central limit theory, extreme value theory (EVT) treats fluctuation of sample maxima. EVT is especially useful to model tails probability distribution to the quantiles even higher than the most extreme observation. It provides a robust framework to study the tail behavior. The EVT uses Fisher and Tippet theorem to address the form of the limit distribution for properly normalized sample extreme. There have been a little economic and finance literature using extreme value theory recently. Manganelli surveyed the performance of the value at risk models (Manganelli & Engle, 2001). McNeil studied the estimation of the quantile risk for financial time series (McNeil, 1999). Longin presents an application of EVT to compute the VaR of a market position (Longin, 2001). Gencay evaluated the relative performance of VaR models with the daily stock market returns on nine different emerging markets (Gencay & Selcuk, 2004).

The task of multi-assets optimization in a large space of financial market is difficult to perform regardless of using

any of the above mentioned methods. In addition, the threshold setting is another important issue on EVT method. The previous studies use statistical chart (McNeil & Frey, 2000, Oh, Kim, Min, & Lee, 2006) or bootstrap method (Reiss & Thomas, 1997) to estimate the threshold value. If the number of assets of portfolio is large then the time needed to perform a numerical estimate of the efficiency of portfolios composed by assets characterized by different thresholds is too high for real-time model usage.

The genetic algorithm (GA), introduced by Holland in 1975, is a well-known efficient nonlinear search methodology in large spaces (Holland, 1975). Genetic algorithms use a population, which is simply a set of chromosomes, to search the solution space. During each generation, three genetic operators are applied to the population: selection, crossover, and mutation. Selection operator picks chromosomes in the population based on their fitness. Each pair of chromosomes or parents undergo crossover at random by exchanging their information with each other to generate new chromosomes or offspring. The efficient set selections within a portfolio have been efficiently solved by using GA (Shoaf & Foster, 1998; Xia, Liu, Wang, & Lai, 2000; Xia, Wang, & Deng, 2001). These GA-based portfolio selection algorithms only focus on standard deviation as the appropriate measure for risk. This indicates that investors weigh the probabilities of negative returns equally against positive returns (Campbell et al., 2001). In addition, GA also uses to measure portfolio volatility relative the stock market (Oh et al., 2006); evidence the technical trading is efficient in the Santa Fe artificial stock market (Ehrentreich, 2006); explore the relationship between wealth dynamics and risk preferences in the multi-asset artificial stock market (Chen & Huang, 2007). In this paper, risk evaluations concentrate on estimating the shapes of the tail distributions of financial return series.

The objective of this article is to optimize the portfolio set under appropriate threshold selection. Thus we introduce a GA-based portfolio volatility forecasting model to extract the best portfolio set and dynamically estimate a suitable peak threshold for each asset in the portfolio, simultaneously. These peak thresholds are used to estimate the portfolio's VaR by using EVT. In addition, the peak-over-threshold (POT) method is used to collect returns in the series that exceed a certain high threshold.

The remaining part of this paper is organized as follows. Section 2 describes the PVaR using the EVT model under the GA mechanism. In this Section, we introduce encoding method, PVaREVT model and the design of fitness function. Section 3 shows our experimental results. Finally, Section 4 gives our conclusions.

2. PVaR using the EVT model under the GA mechanism

The proposed GA-based PVaR forecasting approach that uses the EVT model depicted in Fig. 2.1. applies the GA mechanism to learn and obtain the best population. In the GA mechanism, the target assets are selected from

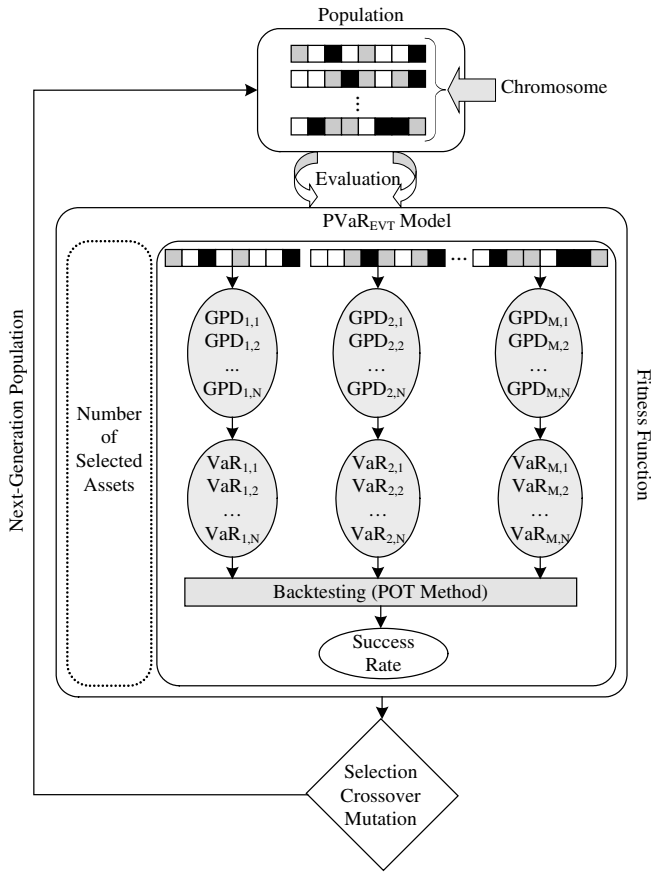


Fig. 2.1. PVaR using the EVT model under the GA Mechanism.

each of the chromosomes (individuals) to be the candidates to forecast the efficient portfolio set in the current population. Then, this mechanism applies GA operations (e.g., selection, crossover, and mutation) to create the next-generation population according to the fitness of the chromosomes. The fitness function focuses on the PVaR evaluation. In the PVaREVT model, the POT method is used to calculate VaR which built the history returns in series that exceed a certain high threshold value and to model these returns separately in GPD distribution. “Backtesting” is used to estimate the success rate of each individual. The number of portfolio sets is also considered in order to arrive at a reasonable portfolio if the fitness function is being designed.

2.1. Genetic algorithm

Genetic algorithm is a search method based on observation of natural evolution and genetics. It combines the survival of fittest with a structured and randomized information exchange mechanism. It is a simple but powerful computation tool and makes no restrictive assumptions about the search space. A basic evolutionary algorithm introduces a simple selection–crossover–mutation–evaluation loop as outlined below (Fig. 2.2). The first step in order to solve a problem by using the GA tool is to design

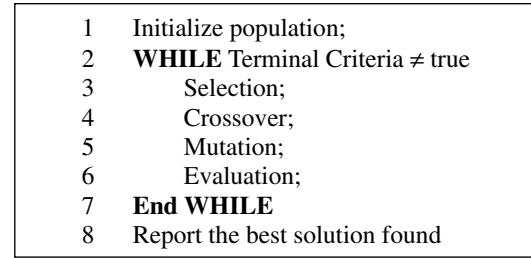


Fig. 2.2. Evolutionary algorithm.

an encoding mechanism which represents each possible solution of the problem by means of a fixed length (binary) string called chromosome. Each chromosome will be evaluated by a fitness function for its goodness.

2.2. Encoding method

In order to represent a set of selected portfolios, let us consider a chromosome divided into n genotypes $\{X_i\} i = 1, n$. Each of them is coded by using a 5 bit string (Fig. 2.3). The first bit represents the presence (1) or absence (0) of the corresponding asset in each genotype. The other 4 bits corresponding to (24) numbers represent the threshold value of each asset. It means that distinct threshold is allowed to be adjusted to reflect real situation for each asset, respectively, because the real thresholds of assets are different. In the past, most researches only considered the same level of threshold in a portfolio for simplicity. The search space in the portfolio volatility forecasting for genetic algorithms is thus $25n$.

2.3. The PVaREVT model

The PVaREVT model estimates the parameters (β_i, ξ_i) of the asymptotic distribution of selected minimal returns from the generalized pareto distribution (GPD), and the probability (p_i) of a minimal return not exceeding a certain high threshold, u_i , in asset X_i . Assume that the collected returns in the series data $\{r_{1,u_i}^j, r_{2,u_i}^j, \dots, r_{N,u_i}^j\}$ are above the threshold u_i which follows the excess distribution $F_{u_i}(x)$ shown in Eq. (2.1).

$$F_{u_i}(x) = P(r^j - u_i \leq x | r^j > u_i) = \frac{F(u_i + x) - F(u_i)}{1 - F(u_i)}$$

$$= \frac{F(r^j) - F(u_i)}{1 - F(u_i)} \quad 0 \leq x \leq \infty, x = r^j - u_i \quad (2.1)$$

$F_{u_i}(x)$ is approximated to GPD as shown in Eq. (2.3), if u_i is high enough for a large class of distributions.

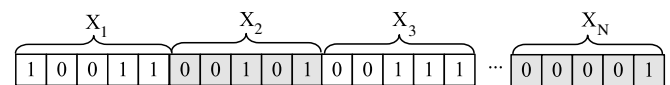


Fig. 2.3. Encoding a portfolio and threshold into chromosomes.

$$\lim_{u \uparrow} \sup_{0 \leq x \leq r^i - u_i} |F_{u_i}(x) - G_{\beta_i, \xi_i}(x)| = 0 \quad (2.2)$$

$$G_{\beta_i, \xi_i}(x) = \begin{cases} 1 - \left(1 + \xi_i \frac{x}{\beta_i}\right)^{-\frac{1}{\xi_i}}, & \xi_i \neq 0 \\ 1 - e^{-\frac{x}{\beta_i}}, & \xi_i = 0 \end{cases} \quad (2.3)$$

The corresponding probability distribution function (PDF) is shown in Eq. (2.4), where β_i is a positive scaling parameter, and ξ_i is the tail index for the fat-tailed distributions found in finance.

$$g_{\beta_i, \xi_i}(x_j) = \frac{1}{\beta_i} \left(1 + \xi_i \frac{x_j}{\beta_i}\right)^{-\frac{1}{\xi_i} - 1} \quad (2.4)$$

We estimate them using maximum likelihood estimation (MLE) as shown below. Let $L(x)$ and $l(x)$ denote the maximum-likelihood function and log-likelihood function, respectively, where $l(x) = \ln(L(x))$.

$$\begin{aligned} L(x) &= g_{\beta_1, \xi_1}(x_1) g_{\beta_2, \xi_2}(x_2) \dots g_{\beta_n, \xi_n}(x_n) \\ &= \left[\frac{1}{\beta_1} \left(1 + \xi_1 \frac{x_1}{\beta_1}\right)^{-\frac{1}{\xi_1} - 1} \right] \times \left[\frac{1}{\beta_2} \left(1 + \xi_2 \frac{x_2}{\beta_2}\right)^{-\frac{1}{\xi_2} - 1} \right] \\ &\quad \times \dots \times \left[\frac{1}{\beta_n} \left(1 + \xi_n \frac{x_n}{\beta_n}\right)^{-\frac{1}{\xi_n} - 1} \right] \end{aligned} \quad (2.5)$$

$$\begin{aligned} l(x) &= -n \ln \beta_i + \left(-\frac{1}{\xi_i} - 1\right) \left[\ln \left(1 + \xi_i \frac{x_1}{\beta_i}\right) \right. \\ &\quad \left. + \ln \left(1 + \xi_i \frac{x_2}{\beta_i}\right) + \dots + \ln \left(1 + \xi_i \frac{x_n}{\beta_i}\right) \right] \\ &= -n \ln \beta_i + \left(-\frac{1}{\xi_i} - 1\right) \sum_{i=1}^n \ln \left(1 + \xi_i \frac{x_i}{\beta_i}\right) \end{aligned} \quad (2.6)$$

To maximize $l(x)$, we take partial differential of $l(x)$ with respect to β_i and ξ_i , respectively. We have

$$\frac{\partial l(x)}{\partial \beta} = -\frac{n}{\beta} - \left(\frac{1}{\xi} + 1\right) \sum_{i=1}^n \frac{-\xi \frac{x_i}{\beta^2}}{1 + \xi \frac{x_i}{\beta}} = 0 \quad (2.7)$$

$$\frac{\partial l(x)}{\partial \beta} = \frac{1}{\beta^2} \sum_{i=1}^n \ln \left(1 + \xi \frac{x_i}{\beta}\right) + \left(-\frac{1}{\xi} - 1\right) \sum_{i=1}^n \frac{x_i}{\beta + \xi x_i} = 0 \quad (2.8)$$

After estimating the GPD distribution and its parameters, the VaR_i of the underlying return distribution $F_{u_i}(x)$ is calculated as in Eq. (2.9) for asset X_i , where $\hat{\beta}_i$ and $\hat{\xi}_i$ are the estimating solution of Eqs. (2.7) and (2.8).

$$\text{VaR}_i = u_i + \frac{\hat{\beta}_i}{\hat{\xi}_i} \left\{ \left[\frac{T}{N_{u_i}} (1 - p_i) \right]^{-\hat{\xi}_i} - 1 \right\} \quad (2.9)$$

2.4. Fitness function design

As a general framework of our proposed fitness function of PVaR uses extreme value theory (EVT) algorithm (Fig. 2.4). The input variables are chromosome set and stock returns for each sample. The chromosome set is generated by selection, crossover and mutation operators. A series of stock returns $\{r_{1,u_i}^i, r_{2,u_i}^i, \dots, r_{N_{u_i},u_i}^i\}$ above distinct threshold u_i are extracted from database for each sample.

In this paper, the higher success rate when less assets are considered in the portfolio is preferred. Based on the recommendations of the basel capital accord (BCA) in 1996, we use the “backtesting” method to evaluate the reliability of our model. Let T denote the length of the observed period in Fig. 2.5.

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6.1 Input: Chromosome set, Stock returns for each sample
6.2 Output:  $\text{VaR}_i$ ,  $\hat{\beta}_i$ ,  $\hat{\xi}_i$ 
6.3   FOR each Chromosome
6.4     Filter and group bits for each sample following our encoding scheme;
6.5     FOR each sample
6.6       IF this sample ( $X_i$ ) is selected THEN
6.7         Get a series of corresponding returns  $\{r_1^i, r_2^i, \dots, r_{N_{u_i}}^i\}$  for  $X_i$ ;
6.8         Sort  $\{r_1^i, r_2^i, \dots, r_{N_{u_i}}^i\}$  in ascending order;
6.9         Extract the returns  $\{r_{1,u_i}^i, r_{2,u_i}^i, \dots, r_{N_{u_i},u_i}^i\}$  above threshold  $u_i$ ;
6.10        calculate  $\hat{\beta}_i$ ,  $\hat{\xi}_i$ ;
6.11        calculate  $\text{VaR}_i$ ;
6.12        IF  $\text{VaR}_i$  large than real daily return THEN
6.13          Success;
6.14        ELSE
6.15          Failure;
6.16        End IF
6.17      End IF
6.18    End FOR
6.19    Calculate the fitness of this portfolio by using the average success rate;
6.20  End FOR

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Fig. 2.4. Fitness function of PVaR uses extreme value theory (EVT) algorithm.

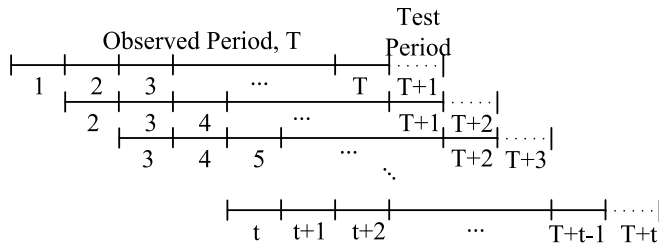


Fig. 2.5. Sliding window simulation process.

Each window consists of T observed values and one test value. The overall simulation process uses the sliding window methodology, which gradually shifts one time slot until the end of our experiment. By substituting these T observed values into Eqs. 2.3, 2.4, 2.5, 2.6, 2.7, 2.8, 2.9, the VaR_i is calculated, where $\hat{\beta}_i$ and $\hat{\xi}_i$ are estimated by using Newton's method. By comparing VaR_i with actual return data based on test values, we observe that success is achieved if VaR_i is larger, otherwise, failure results. After shifting the sliding window t times, the success rate is defined as follows:

$$\text{Success rate} = \frac{\text{Number of success}}{t}. \quad (2.10)$$

In our experiment, T is set to 832, because there are totally 932 observed periods from 1987 to 2004, and the sliding window shifts 100 times ($t = 100$). It means that there are 832 observations in the observed period (OP) used to train our model and the next one observation in the test period (TP) used to test the performance of our model. Let OP_i and TP_i denote the Observed Period and Test Period in the i -th window, respectively. $(OP_1, TP_1) = (1-832, 833)$, $(OP_2, TP_2) = (2-833, 834), \dots$, and $(OP_{100}, TP_{100}) = (100-931, 932)$. Finally, the average success rate would be got easily in this portfolio.

Thus, both the success rate criterion and the number of assets criterion are considered simultaneously. Let $F(V)$ denote the fitness function which is defined as follows, where SR_{VaR} denotes the success rate calculated by the PVaREVT model, and S represents the number of selected assets:

$$F(V) = \frac{SR_{VaR}}{(|N| - |S|)/|N|} \quad (2.11)$$

3. Experimental results

Simulation environment, sample data and experimental results are described in this section. Our PVaR using the EVT model under the GA program is written in Borland C++ Builder 6.0.

3.1. GA parameters

The parameters used in our GA runs are set as follows. The population size is 500 and chromosome length is 390

since there are 78 assets each of them represented by 5 bits. The number of generations is set to 1000. The selection method is roulette wheel and the crossover method is one-point crossover. The crossover rate is set to 0.6 and mutation rate is set to 0.001 as suggested in Srinivas and Lalit (1994). Minor changes in these parameters did not seem to have a major effect on the performance in our preliminary tries.

3.2. Sample data

There are more than 700 companies currently listed in Taiwan stock exchange (TSE). We remove companies which are listed less than 20 years in Taiwan economic journal (TEJ) data bank. The results analyses are 20 industries of 78 companies shown in Appendix. We use weekly stock return data because 7% daily price upward/downward limited in TSE. There are 932 observations in data set of each company from January 1987 to December 2004.

The level of the TSE index at weekly returns is presented in top panel of Fig. 3.1. The weekly TSE index is selected from January first week 1987 to December last week

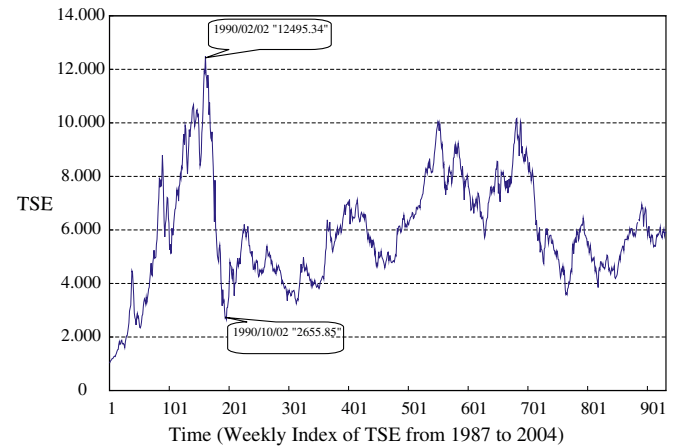


Fig. 3.1. The weekly Index of TSE Index from 1987 to 2004.

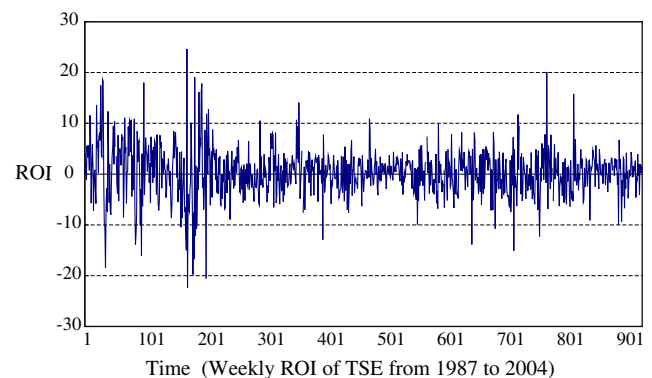


Fig. 3.2. The weekly return of TSE Index from 1987 to 2004.

2004. The horizontal axis corresponds to time whilst the vertical axis displays the value of index. The corresponding weekly returns are displayed in the bottom panel of Fig. 3.2. The TSE index returns calculated by $rt = (pt + 1 - pt)/pt$, where pt is the value of the index at week t . However, extremely weekly return is as high as 24.64% or weekly losses are as low as -22.39% that are observed during the sample period. The average weekly return is 0.2959% which implies approximately 16.60% annual return (52 business weeks) based on $1.00295952 = 1.166$. Furthermore, 4.97% weekly standard deviation indicates a highly volatile environment. The result is not surprising because the economy of Taiwan experienced a violent oscillation in inflation, especially since 1986 to the early 1990. The TSE index in this period shot up more than 10-fold and then plummeted by more than 75% from the peak in just 6 months. Such large fluctuations of economy of Taiwan have often been considered as the boom and burst of a bubble.

The descriptive statistics of those companies are given in Table 3.1 which uses year 2004 TEJ data unless otherwise specified. Some interesting remarks about the characteristics of these companies can be resumed into three points. First, the mean net sales is \$2045.28 million (with 9 companies greater than mean) and the mean total assets is \$63259.05 million (with 13 companies greater than mean) which indicate that the net sales and total assets of most sample companies are less than their respective mean values. This data is caused by the fact that most of the selected companies belong to the small and middle enterprises. Second, the mean market value of equity \$28348.44 is significantly larger than net sales, and both the mean leverage (0.72) and current ratio (2.32) fall into normal range. Third, the low sales growth rate reflects the current global depression. Finally, these target companies possess greater profit, but they appear higher risk of daily turnover because their mean are 2.44 and 8.40, respectively.

Table 3.1
The descriptive statistics of sample companies

Items	Mean	Number > mean	Maximum	Minimum
Net sales (millions)	\$2045.28	9	\$58016.75	\$-1416.38
Total assets (millions)	\$63259.05	13	\$1313155.97	\$699.57
Market value of equity (millions)	\$28348.44	3	\$363830.78	\$77.00
Sales growth rate (%)	1.63	36	103.13	-68.07
Return on total assets (%)	2.44	40	22.82	-36.72
Current ratio (%)	2.32	18	2753.95	9.63
Leverage (%)	0.72	56	6.29	-10.11
Turnover (%)	8.40	29	3.73	0.01

3.3. Results

In this study, we proposed GA to optimize an efficient portfolio set and threshold value of EVT model in order to improve the difficult of volatility forecasting of multivariate assets of VaR. To show the effectiveness of GA on PVaR using EVT, and the experimental results are compared with Historical Simulation and Exponential Weighted Moving Average as follows:

- GA-PVaR-HS (GA-based portfolio VaR forecast with historical simulation method): HS works as $\text{VaR}_t(\alpha) = F^{-1}(\alpha)$, where $F^{-1}(\alpha)$ is the q -th quantile ($q = 1 - \alpha$) of the sample distribution. The portfolio returns are sorted in ascending order and the θ -quantiles of interest is given by the return that leaves $\theta\%$ of the observations on its left side and $(1 - \theta)\%$ on its right side.
- GA-PVaR-EWMA (GA-based portfolio VaR forecast with exponential weighted moving average method): The EWMA method can be alternatively written as $\sigma_{t+1}^2 = (1 - \lambda) \sum \lambda^{i-1} (r_{t+1-i} - \bar{r}_t)^2$ where λ is the decay factor $0 < \lambda < 1$. The EWMA can be seen as an infinite weighted average of past squared returns but with exponentially declining weights. It can be interpreted as an infinite order autoregressive model for the squared return.
- GA-PVaR-EVT (GA-based portfolio VaR forecast with extreme value theory method): its algorithm was illustrated in Section 2.3.

We divide the empirical result analysis into two parts: the PVaR forecasting results with these three models is proposed in Section 3.3.1 and the detailed parameters analyses of individual assets of VaR in portfolios using GA-PVaR-EVT method are shown in Section 3.3.2.

3.3.1. PVaR forecasting analysis of different model

The PVaR of TSE using HS, EWMA and EVT methods with 95% and 99% confidence level are listed in Tables 3.2 and 3.3, respectively. In Table 3.2, the PVaR of GA-PVaR-HS, GA-PVaR-EWMA and GA-PVaR-EVT are 10.52, 9.58 and 16.97, respectively. The success rate of GA-PVaR-HS, GA-PVaR-EWMA and GA-PVaR-EVT are 92.48%, 90.38% and 99.71%, respectively. The success rate of GA-PVaR-EVT 99.71% is better than confidence level 95%. On the contrary, the success rates of GA-PVaR-HS, GA-PVaR-EWMA are smaller than the confidence level. The GA-PVaR-HS, GA-PVaR-EWMA is thus obviously underestimated than the GA-PVaR-EVT. It means that our model is overvalued and GA-PVaR-HS, GA-PVaR-EWMA are undervalued.

The similar results happen in Table 3.3. The PVaR of GA-PVaR-HS, GA-PVaR-EWMA and GA-PVaR-EVT with 99% confidence level are 19.18, 9.58 and 16.97, respectively. The success rate of GA-PVaR-HS, GA-

Table 3.2

PVaR of TSE using HS, EWMA and EVT methods with 95% confidence level

GA-PVaR-HS			GA-PVaR-EWMA			GA-PVaR-EVT		
StockID	Success rate (%)	VaR	StockID	Success rate (%)	VaR	StockID	Success rate (%)	VaR
1210	93.48	−8.69	1201	92.73	−11.97	1207	99.24	−33.14
1414	92.73	−13.76	1207	88.94	−11.49	1229	99.24	−39.00
1419	89.23	−11.40	1416	91.21	−12.34	1305	99.24	−54.30
1434	92.73	−8.60	1423	88.18	−7.96	1409	99.24	−36.29
1503	93.48	−8.78	1704	90.45	−8.01	1410	100.00	−57.71
1506	93.48	−12.56	1904	90.45	−8.44	1418	100.00	−38.28
1603	91.21	−11.96	2103	89.70	−10.57	1419	100.00	−38.04
1701	92.73	−11.96	2704	88.94	−6.19	1423	100.00	−22.61
1707	94.24	−10.42	2812	93.48	−9.71	1434	100.00	−39.38
1708	92.73	−11.29	2903	89.70	−9.11	1503	100.00	−25.66
1709	93.48	−9.07				1601	99.24	−34.85
1718	91.21	−12.63				1604	99.24	−37.94
2002	94.24	−7.77				1605	98.48	−24.40
2808	91.97	−9.87				1704	100.00	−57.46
2915	90.25	−9.04				1802	100.00	−15.89
						2101	100.00	−26.71
						2102	100.00	−48.55
						2303	100.00	−13.51
						2601	100.00	−50.60
						2901	100.00	−35.88
						2915	100.00	−24.94
Mean	92.48	10.52		90.38	9.58		99.71	16.97

Table 3.3

PVaR of TSE using HS, EWMA and EVT methods with 99% confidence level

GA-PVaR-HS			GA-PVaR-EWMA			GA-PVaR-EVT		
StockID	Success rate (%)	VaR	StockID	Success rate (%)	VaR	StockID	Success rate (%)	VaR
1210	98.56	16.78	1201	98.48	17.05	1104	100.00	13.14
1213	99.62	23.13	1207	95.28	16.76	1212	100.00	13.00
1229	99.24	18.95	1409	95.48	18.97	1307	100.00	34.30
1419	96.00	18.47	1416	99.23	17.89	1308	100.00	16.29
1423	93.58	22.26	1418	96.21	13.58	1413	100.00	17.71
1432	94.36	17.76	1704	97.73	11.6	1414	100.00	18.28
1434	97.26	13.34	1904	96.97	12.12	1416	100.00	18.04
1503	99.85	18.62	2103	97.73	15.41	1503	100.00	2.61
1506	98.12	24.92	2201	98.48	12.58	1601	100.00	19.38
1707	95.00	20.22	2303	95.26	12.13	1603	100.00	5.66
1709	96.26	16.76	2704	96.12	9.12	1606	100.00	14.85
1905	97.73	16.99	2808	97.12	9.87	1701	100.00	17.94
1909	99.24	19.5	2809	96.48	13.6	1704	100.00	4.40
2002	98.26	14.07	2812	99.24	13.82	1708	100.00	27.46
2101	97.15	27.79	2903	97.73	13.18	1709	100.00	5.89
2303	98.23	16.98				1718	100.00	6.71
2601	95.36	22.62				1802	100.00	28.55
2801	94.23	16.4				2002	100.00	3.51
2901	99.23	22.42				2201	100.00	20.60
2915	94.28	15.62				2506	100.00	15.88
						2701	100.00	4.94
						2807	100.00	34.01
						2812	100.00	24.10
						2901	100.00	2.08
						2903	100.00	30.36
						2913	100.00	18.00
Mean	97.08	19.18		97.17	13.85		100.00	16.07

PVaR-EWMA and GA-PVaR-EVT are 97.08%, 97.17% and 100.00%, respectively. It is obvious that the success

rate of GA-PVaR-EVT is larger than its confidence level 99%.

Table 3.4

The parameters of individual stock with 99% confidence level (GA–PVaR–EVT)

Stock ID	Success rate (%)	u (%)	VaR	β	ξ
1104	100.00	7.00	13.14	0.48	0.0029
1212	100.00	7.00	13.00	0.65	0.0043
1307	100.00	6.00	34.30	0.55	0.0021
1308	100.00	7.00	16.29	0.35	0.0011
1413	100.00	6.00	17.71	0.86	0.0009
1414	100.00	7.00	18.28	0.17	0.0016
1416	100.00	7.00	18.04	0.88	0.0029
1503	100.00	8.00	2.61	0.02	0.0004
1601	100.00	7.00	19.38	0.54	0.0051
1603	100.00	8.00	5.66	0.4	0.0030
1606	100.00	7.00	14.85	0.07	0.0009
1701	100.00	7.00	17.94	0.6	0.0016
1704	100.00	8.00	4.40	0.05	0.0003
1708	100.00	6.00	27.46	0.98	0.0011
1709	100.00	9.00	5.89	1.24	0.0010
1718	100.00	8.00	6.71	0.27	0.0007
1802	100.00	6.00	28.55	0.44	0.0017
2002	100.00	9.00	3.51	0.58	0.0024
2201	100.00	6.00	20.60	0.02	0.0023
2506	100.00	7.00	15.88	0.36	0.0032
2701	100.00	8.00	4.94	1.85	0.0012
2807	100.00	6.00	34.01	0.31	0.0055
2812	100.00	6.00	24.10	0.38	0.0001
2901	100.00	10.00	2.08	4.76	0.0011
2903	100.00	6.00	30.36	0.09	0.0007
2913	100.00	7.00	18.00	0.79	0.0043
Mean	100.00	7.15	16.07	0.68	0.0020

3.3.2. Parameter analysis of GA–PVaR–EVT models

The POT method deeply depends on the selection of threshold. The asymptotic theory will break down while the threshold value is low enough. Inversely, the number of samples used to estimate the parameters in the excess distribution would be insufficient if the threshold value is too high. Table 3.4 shows the parameters calculated using GA–PVaR–EVT of POT method with 99%. The mean of success rate is 100%. The mean of threshold value u is 7.15% which follows the recommendations (5–10% of the sample) from the simulation study in McNeil and Frey (2000). The tail index ξ and scaling parameter β are determined by fitting the GPD to the real data. We find the GPD-based method provides much better estimates of market risk than historical simulation and exponential weighted moving average method, because it can deal with asymmetries in the fat-tailed distribution with $\xi > 0$ and $\beta > 0$.

4. Conclusion

Forecasting the volatility of a portfolio is important in investment decisions. Conventionally, forecasting PVaR and selecting an efficient set of portfolios should have serious limitations if statistical linear models based on specific assumptions are used. Traditional risk management models might fail to forecast VaR precisely because the tail distributions are estimated incorrectly. Most real financial time series data comprise fat-tailed distributions. The EVT model has been proved to be a powerful approach to risk

forecasting because it focuses directly on the distribution of extreme returns instead of the whole distribution of returns. However, forecasting the PVaR is difficult because EVT provides less maturity when analyzing the multivariate model. In this article, a GA-based PVaR forecasting mechanism using the EVT model is introduced, where the GA mechanism extracts an efficient set of portfolios, and simultaneously estimates more suitable threshold values for all assets in order to forecast PVaR more precisely. Using 78 companies listed and traded on the TSE, we try to demonstrate the stability and robustness of our mechanism. The experimental results show that the success rates of GA-based PVaR with extreme value theory are higher than EWMA and HS methods both with 95% or 99% confidence level. It implies that higher success rate can effectively avoid undervalued market risk of portfolio. This result is important in financial markets because undervalued risk will lead to possible financial distress or bankruptcy. Besides, the GA-based PVaR forecasting mechanism can efficiently select more suitable thresholds for each asset in portfolio. GPD-based approach to estimation tail distribution in this paper is more efficient than historical simulation and exponential weighted moving average method. The value of tail index ξ and scaling parameter β in our results are approximately to be reasonable since they can effectively deal with asymmetries in the fat-tailed distribution.

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Appendix

Industries	Stock ID	Business name
Cement	1101	Taiwan Cement
	1102	Asia Cement
	1103	Chia Hsin Cement
	1104	Universal Cement
Food	1201	Wei Chuan Food
	1207	Chia Hsin
	1210	Great Wall Enterprise
	1212	Sino Japan International
	1213	Oceanic Beverage
	1229	Lien Hwa Ind.
Plastic and Petroleum	1301	Formosa Plastics
	1303	Nan Ya Plastic
	1305	China General Plastics
	1307	San Fang Chemical
	1308	Asia PolyMer
	1309	Taita Chemical

Appendix (continued)

Industries	Stock ID	Business name
Textile and Fiber	1402	Far Eastern Textile
	1408	Chung Shing Textile
	1409	Shinkong Synthetic Fibers
	1410	Nan Yang Dyeing and Finishing
	1413	Hung Chou Chemical
	1414	Tung Ho Textile
	1416	Kwong Fong Industries
	1417	Carnival
	1418	Tong Hwa Synthetic Fiber
	1419	Sinkong Spinning
	1423	Reward Wool
	1432	Taroko Textile
	1434	Formosa Taffeta
Electrical Machinery	1503	Shihlin Elec. and Eng.
	1504	Teco Elec. and Mach
	1506	Right Way Ind.
Wire and Cable	1601	Taiwan Fluorescent
	1603	China Wire and Cable
	1605	Walsin Lihwa
	1606	Kolin
Chemical	1701	China Chemical
	1702	Namchow Chemical
	1704	Lee Chang Yung Chemical
	1707	GRAPE KING
	1708	SESoda
	1709	Formosan Union Chemical
Glass	1718	China Man-Made Fiber
	1802	Taiwan Glass
Pulp and Paper	1902	Taiwan Paper
	1904	Cheng Loong
	1905	Chung Hwa Pulp
	1907	Yuen Foong Yu Paper
	1909	Long Chen Paper
Iron and Steel	2002	China Steel
Rubber	2101	Nan Kang Rubber Tire
	2102	Federal
	2103	Taiwan Synthetic Rubber
	2104	China Synthetic Rubber
Automobile	2201	YULON MOTOR
Electronics	2302	Rectron
	2303	United Micro Electronics
	2371	Tatung
	1604	Sampo
Construction	2501	Cathay Real Estate Deve.
	2504	Goldsun Devel. and Const.
	2506	Pacific Construction

Appendix (continued)

Industries	Stock ID	Business name
Transportation	2601	First Steamship Company
Tourism	2701	Wan Hwa Enterprise
	2704	Ambassador Hotel
Banking and Insurance	2801	Chang Hwa Bank
	2807	Hsinchu International Bank
	2808	Int'l Bank of Taipei
	2809	The MB Bank of Taiwan
	2811	Taitung Business Bank
	2812	Taichung Business Bank
Retailing	2901	Shin Shin Supermarket
	2903	Far East Dept.
	2904	Pan Overseas
	2913	Taiwan Tea
	2915	Ruentex Industries.Limited
Conglomerates	9801	China Rebar
Management	1458	Chia Hsin Livestock

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